

Hands on Homework #6 (Truss)

EGN 3310 - 0M01 Engineering Analysis: STATICS



MWF, 11:30AM-12:20PM

Submitted by: Group 38

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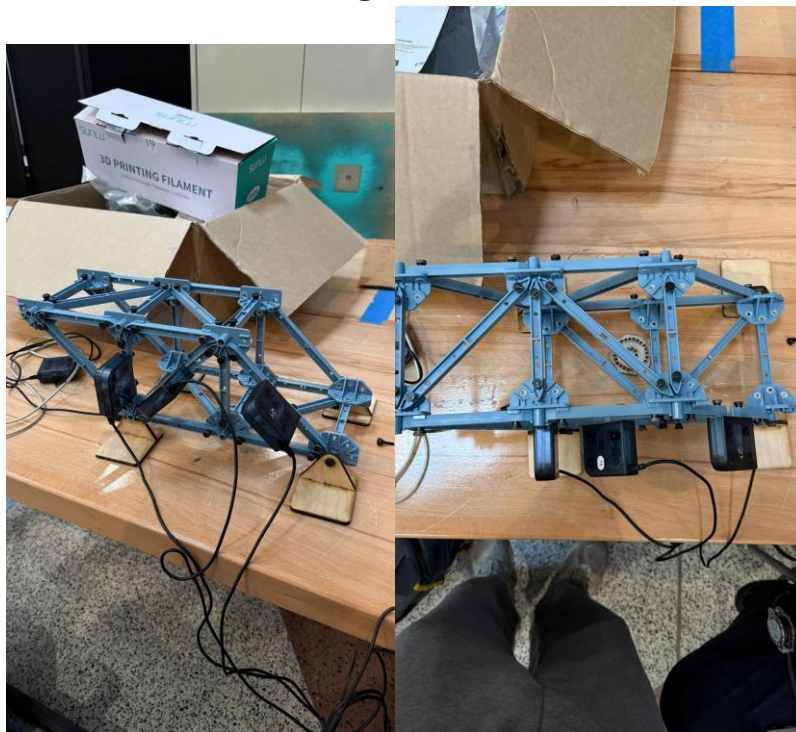
*** Before you read through, we wanted to let you know that we did mess up with the ending NID number. We thought one of ours ended on a 9 and ended up doing the last one by accident.

We realized this at the end of everything before submitting on Friday, and we went home so we could not recalculate everything.

We are truly sorry, please don't be too harsh on us.

Hands-On Method (Physical Model)

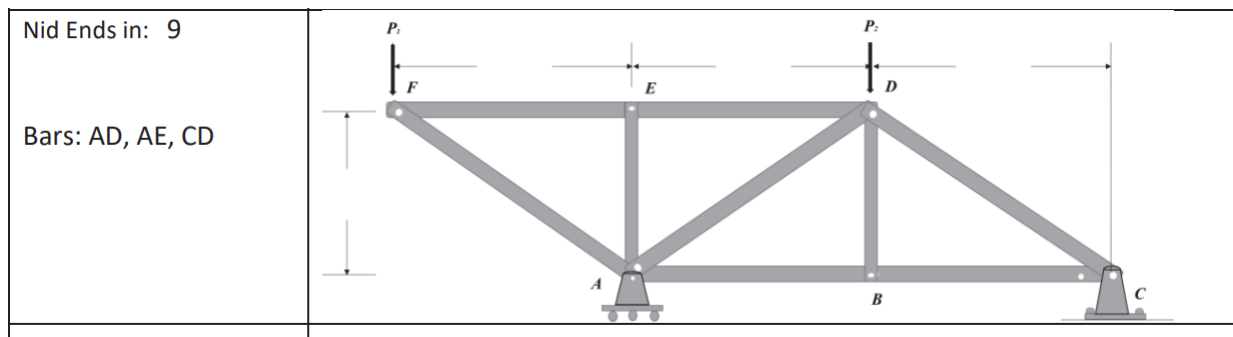
Photos of our bridge



Photos of our bridge with mass on it



As you can see, we completed a one to one 3D model of the TRUSS that ends with the UCF ID 9.



After building the model, we applied the sensors on the required bars and applied weights for both P1 and P2.

$$P1 = 500g = 4.905N$$

$$P2 = 200g = 1.47N$$

Upon placing these, we got the following values for the bars:

	Run #2	Run #2	Run #2
	Force 1 (N)	Force 2 (N)	Force 3 (N)
1	1.94	0.41	0.25
2	1.93	0.39	0.27
3	1.95	0.41	0.25
4	1.93	0.39	0.25
5	1.93	0.40	0.25
6	1.92	0.40	0.24
7	1.93	0.38	0.24
8	1.92	0.40	0.26
9	1.92	0.40	0.24
10	1.93	0.40	0.25
11			

Taking the averages for each column we get:

Force 1 (AD) = 1.93N

Force 2 (CD) = 0.398N

Force 3 (AE) = 0.25 N

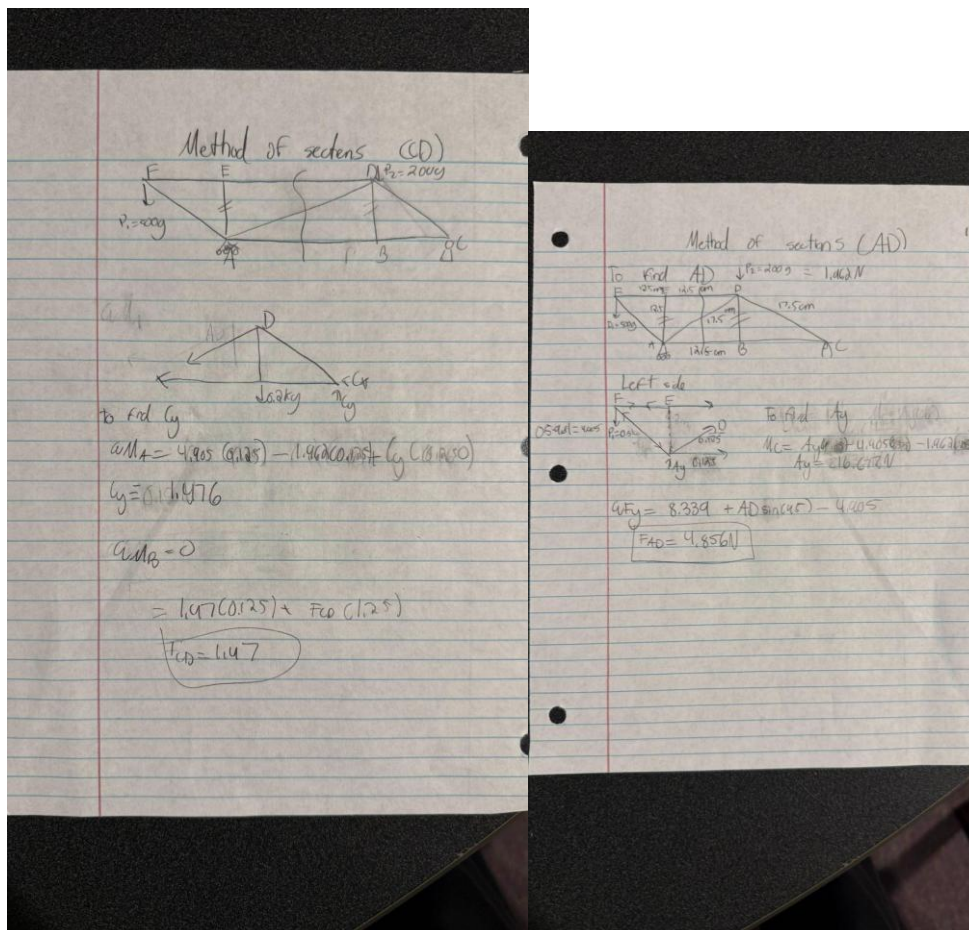
That concludes the hands-on section, and this was really fun to build.

Hand Calculation (Theoretical Method)

Method of sections

We started off with the method of sections.

Here is the paper with all of our calculations:



We immediately were able to eliminate AE and BD as they are zero force members. This is the case because each has 2 collinear bars and they are the third one left out.

Now in terms of calculations, we'll go through AD first and then CD.

For AD, we first started out by finding A_y from the roller, as this is a key component to later set the forces in the y direction = 0

$$M_C = 0$$

$$0 = A_y(0.125) - 4.905(0.375) - 1.96(0.125)$$

Solving this we get that A_y is 16.677N

Following this to find AD we set the $F_y = 0$, then solve for AD

$$F_y = 8.339 + AD \sin(45^\circ) - 4.905$$

$$AD = 4.856N$$

For CD, we first started out by finding C_y from the pin, once again by using the moment. However this time we used the moment from joint A. For now we'll go over this formula.

$$M_a = 4.905(0.125) - 1.962(0.125) + C_y(0.250)$$

This leads to C_y being 1.476N

This is where we deviated a little bit, as if we used the sum of forces our percent error was abnormally high.

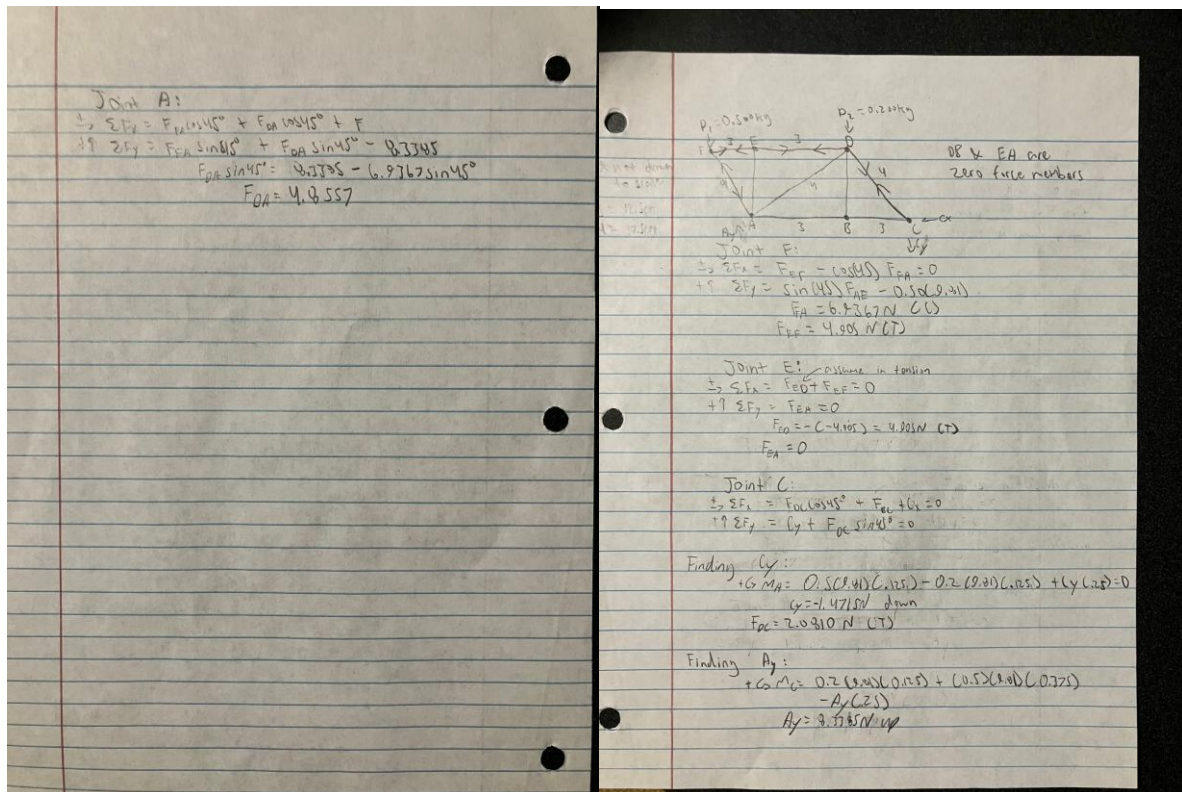
We inspired ourselves from the Zaurin video, and used the summation of moments from B.

$$M_b = 1.476(0.125) + F_{cd}(0.125)$$

This leads to F_{cd} being 1.47N

Method of joints

Here is the paper with all of our calculations:



Calculations

Known Facts: DB and EA are zero force members

Notation: members labelled 3 are 12.5 centimeters long and members labelled 4 are 17.5 centimeters

Joint F:

$$F_x = F_{FE} - \cos(45) * F_{FA} = 0$$

$$F_y = F_{FA} * \sin(45) - 0.50(9.81) = 0$$

$$F_{FA} = 6.9367 \text{ N (C)}$$

$$F_{FE} = 4.905 \text{ N (T)}$$

Joint E:

$$F_x = F_{ED} + F_{EF} = 0$$

$$F_y = F_{EA} = 0$$

$$F_{ED} = 4.905 \text{ N (T)}$$

$$F_{EA} = 0$$

Joint C:

$$F_x = F_{DC} * \cos(45) + F_{BC} + C_x = 0$$

$$F_y = F_{DC} \cdot \sin(45) + C_y = 0$$

Finding C_y and F_{DC} :

$$M_A = 0.5 \cdot 9.81 \cdot .125 - 0.2 \cdot 9.81 \cdot .125 + C_y \cdot (.25) = 0$$

$$C_y = -1.4715\text{N (down)}$$

$$F_{DC} = 2.0810\text{N (T)}$$

Finding A_y :

$$M_A = 0.2 \cdot 9.81 \cdot .125 + .5 \cdot 9.81 \cdot .375 - A_y \cdot .25 = 0$$

$$A_y = 8.3385\text{N (up)}$$

Joint A:

$$F_x = F_{FA} \cdot \cos(45) + F_{DA} \cdot \cos(45) + F_{AB} = 0$$

$$F_y = F_{FA} \cdot \sin(45) + F_{DA} \cdot \sin(45) - A_y = 0$$

$$F_{DA} \cdot \sin(45) = 8.3385 - 6.9367 \cdot \sin(45)$$

$$F_{DA} = 4.8557\text{N}$$

Percent errors

Here is a quick table to recap on the values:

	Model Values	Hand Calculated Values (Sections)	Hand Calculated Values (Joints)
AD	1.93N	4.856N	4.8557N
AE	0.25N	0N	0N
CD	0.398N	1.47N	2.0810N

$$\text{Percent Error} = \frac{|measured - calculated|}{measured} \cdot 100\%$$

For the Methods of Sections:

$$AD = \left| \frac{1.93 - 4.856}{1.93} \right| \cdot 100\% = 151.6\%$$

$$AE = \left| \frac{0.25 - 0}{0.25} \right| \cdot 100\% = 100\%$$

$$CD = \left| \frac{0.398 - 1.47}{0.398} \right| \cdot 100\% = 269.3\%$$

For the Methods of Joints:

$$AD = \left| \frac{1.93 - 4.8557}{1.93} \right| \cdot 100\% = 151.6\%$$

$$AE = \left| \frac{0.25 - 0}{0.25} \right| \cdot 100\% = 100\%$$

$$CD = \left| \frac{0.398 - 2.081}{0.398} \right| \cdot 100\% = 422.9\%$$

It is apparent that these errors are abysmal, and we have a couple of theories for why this might be.

First the obvious one is the sensors having issues. This is the most likely one as AE had a reading of 0.25N when it is a zero force member. We're not really sure how often this gets calibrated, and we are confident in the way we built the bridge. Therefore the readings might have been wrong. Following that we might not have placed the bridge on the supports the way it is intended, as this could make it so the force is not propagated the way it should be.

Conclusion/discussion:

Vincent Tremblay (vi474918):

Overall, I believe we did an okay job. It's hard for us to say if the large error comes from us or not, but based on what people have been saying it leans towards the sensors being uncalibrated.

We were able to successfully calculate using both methods, and apply them to our work.

Overall, it was a great learning experience and a great way to apply statics concepts.

Andrew Tran (an365172):

Despite our large margin of error, I believe that the hands on provides a good visual for how we can represent 3d problems into 2d visual Free Body Diagrams (FBD's) and use past knowledge (summation of forces = 0, summation of moments = 0) to find out the forces on the members, and if they are in compression or tension. If we were to do this experiment again, we would use several different electronic sensors to measure the forces, since the data received from our electronic sensors was very far from what we actually calculated.

Kevin Sanchez Ramos (ke987766):

Based on our experience and knowledge, this was a great opportunity to understand and apply both the method of sections and the method of joints.

Though our hands on calculations results were different, it really helped us learn more about how we can improve in learning more about statistics, as through our calculations, we were able to determine the easier path for solving particular problems and comparing them to the actual values, which we noticed the major flaws and how we can improve in the future for not only statics but other courses.